

SAT Math

Area and Volume 2

Question #	ID		Answer
2.1	f67e4efc	A right circular cylinder has a volume of 45π. If the height of the cylinder is 5, what is the radius of the cylinder? A. 3 B. 4.5 C. 9 D. 40	A
2.2	5afbd8e	What is the length of one side of a square that has the same area as a circle with radius 2 ? A. 2 B. $\sqrt{2\pi}$ C. $2\sqrt{\pi}$ D. 2π	C
2.3	ec5d4823	What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?	360
2.4	151eda3c	A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B? A. $16,000\pi$ B. $20,000\pi$ C. $25,000\pi$ D. $31,250\pi$	B

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Question # ID

2.5 38517165

Answer

31

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

2.6 08b7a3f5

27

A triangular prism has a height of 8 centimeters (cm) and a volume of 216 cm^3 . What is the area, in cm^2 , of the base of the prism? (The volume of a triangular prism is equal to Bh , where B is the area of the base and h is the height of the prism.)

2.7 a2e76b60

C

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 , and the height of the can is 10 cm. If 110 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

A. 7.5 cm^3

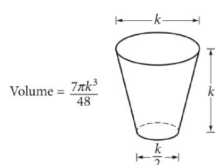
B. 185 cm^3

C. 640 cm^3

D. 750 cm^3

2.8 37dde49f

D



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k , in centimeters?

A. 2.52

B. 7.67

C. 7.79

D. 10.11

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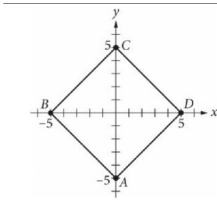
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Question # ID

2.9 cf53cb56

Answer

C



In the xy -plane shown, square $ABCD$ has its diagonals on the x - and y -axes. What is the area, in square units, of the square?

- A. 20
- B. 25
- C. 50
- D. 100

2.10 d621cffb

D

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

2.11 3b931fb0

29

A right circular cylinder has a volume of 377 cubic centimeters. The area of the base of the cylinder is 13 square centimeters. What is the height, in centimeters, of the cylinder?

2.12 e0874bc2

D

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of $\overline{XY}$?

- A. 2
- B. 18
- C. 55
- D. 162

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Question #	ID		Answer
2.13	c0586eb5	A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder? A. 16π B. 96π C. 192π D. 768π	C
2.14	e336a1d2	A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube? A. 164 B. 1,681 C. 10,086 D. 68,921	D
2.15	468613c0	A triangle has a base length of 56 centimeters and a height of 112 centimeters. What is the area, in square centimeters, of the triangle? A. 168 B. 1,568 C. 3,136 D. 6,272	C
2.16	aef4fd8a	The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2, of the square? A. $2 \cdot 94$ B. $2 \cdot 94 \cdot 94$ C. $4 \cdot 94$ D. $94 \cdot 94$	D

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Question # ID

2.17 9c0a0eca

Answer

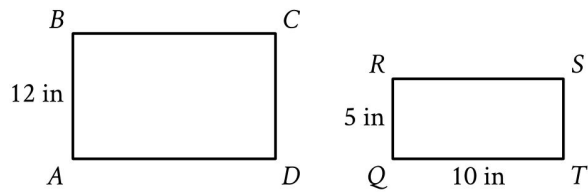
B

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

- A. 700
- B. 350
- C. 175
- D. 80

2.18 e9c5bfb2

B



Note: Figure not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A , B , C , and D correspond to Q , R , S , and T , respectively. What is the length, in inches (in), of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10

2.19 a2659088

D

A right circular cylinder has a height of 8 meters (m) and a base with a radius of 12 m. What is the volume, in m^3 , of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

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Question # ID
2.20 03c6994f

Answer

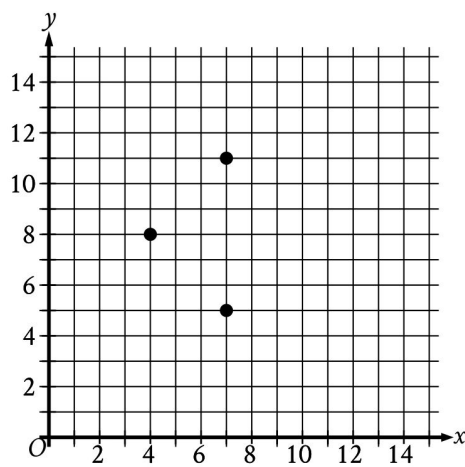
A

Square A has side lengths that are 246 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

- A. 60,516
- B. 492
- C. 246
- D. 123

2.21 096c7ef5

B



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

- A. 3
- B. 6
- C. 7
- D. 9